

Drift Computation Framework

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines how semantic drift is computed from aligned text-unit vectors using the current InterLink architecture. It is the computational core of the experiment and supports all downstream indices, thresholding, null modeling, and interpretability outputs.

Computation Scope

Drift is computed only for units in the shared canon comparison set that are structurally aligned and representation-eligible. This produces two parallel drift universes: R0 using `v_text_raw` and R1 using `v_text_norm`. Each universe is computed independently.

Pair Definition

For each aligned `unit_key`, the study constructs a KJV 1611 unit and a KJV 1769 unit. R0 pairs compare `v_text_raw` from KJV 1611 against `v_text_raw` from KJV 1769. R1 pairs compare `v_text_norm` from KJV 1611 against `v_text_norm` from KJV 1769. Cross-representation comparisons are not used in the primary drift computation.

Drift Metric

For a given aligned unit, drift is defined as cosine distance between the two representation specific vectors. Lower values indicate greater semantic stability and higher values indicate greater measured drift.

- $\text{drift}_i^{R0} = 1 - \text{cosine}(v_{i,1611}^{R0}, v_{i,1769}^{R0})$
- $\text{drift}_i^{R1} = 1 - \text{cosine}(v_{i,1611}^{R1}, v_{i,1769}^{R1})$

Computation Ordering

1. Determine structurally aligned units.
2. Determine representation eligibility for R0 and R1.
3. Retrieve the required named vectors.
4. Compute representation specific drift scores.
5. Write a unified drift record with eligibility flags and nulls where applicable.

Representation Coupling Rule

The framework supports independent representation analysis and matched sensitivity analysis. Independent analysis evaluates all eligible R0 and R1 pairs separately. Matched sensitivity analysis restricts calculations to units where both `drift_r0` and `drift_r1` exist.

Structural Null Handling

If a structurally aligned unit lacks one representation, the unified record is retained with the missing drift value marked null and the eligibility flags preserved. This preserves auditability and enables clean reconciliation with coverage outputs.

Study Contract

Component	Study Decision
Unit of drift	Verse-level aligned pair
R0 vector	v_text_raw
R1 vector	v_text_norm
Distance metric	Cosine distance
Core output	drift_pairs.csv and drift_pairs.parquet
Group hierarchy	Shared canon, OT shared, NT shared, work-level

Expected Output Artifacts

- drift_pairs.csv
- drift_pairs.parquet
- canonical_analysis_records.csv